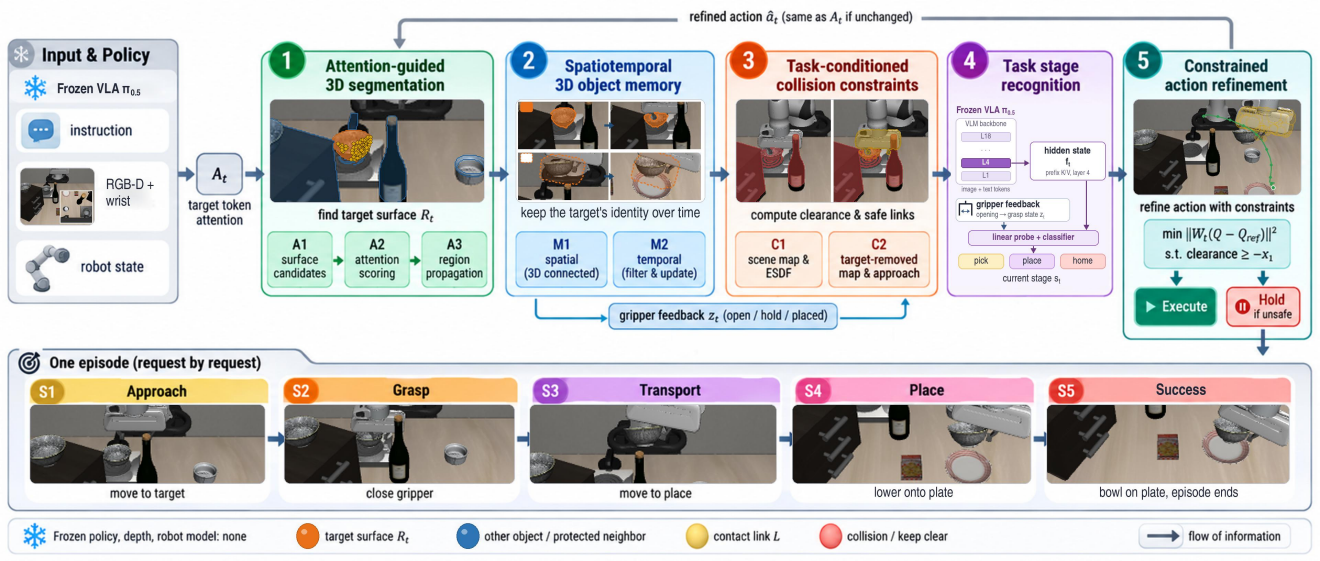

refined action $\hat{a}_t$ (same as A_t if unchanged)
Input & Policy
Frozen VLA $\pi_{0.5}$
instruction
RGB-D + wrist
robot state
A_t
target token attention
1
Attention-guided 3D segmentation
find target surface R_t
A1 surface candidates
A2 attention scoring
A3 region propagation
2
Spatiotemporal 3D object memory
keep the target's identity over time
M1 spatial (3D connected)
M2 temporal (filter & update)
3
Task-conditioned collision constraints
compute clearance & safe links
C1 scene map & ESDF
C2 target-removed map & approach
4
Task stage recognition
Frozen VLA $\pi_{0.5}$
VLM backbone
L18
L4
L1
hidden state f_t
prefix K/V, layer 4
image + text tokens
gripper feedback
opening → grasp state z_t
linear probe + classifier
pick
place
home
current stage s_t
5
Constrained action refinement
refine action with constraints
$\min \|W_t(Q - Q_{ref})\|^2$
s.t. clearance $\geq -x_1$
Execute
Hold if unsafe
gripper feedback z_t (open / hold / placed)
One episode (request by request)
S1 Approach
move to target
S2 Grasp
close gripper
S3 Transport
move to place
S4 Place
lower onto plate
S5 Success
bowl on plate, episode ends
Frozen policy, depth, robot model: none
target surface R_t
other object / protected neighbor
contact link L
collision / keep clear
flow of information